

Hybrid Lichtenberg Algorithm for the Traveling Salesman Problem

A Research Reproduction and Hybrid Extension of a Physics-Inspired Metaheuristic

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Abstract

This report documents the engineering reproduction and combinatorial extension of the **Lichtenberg Algorithm** (LA), a physics-inspired population-based metaheuristic proposed by Pereira et al. (2021) for continuous global optimization. The algorithm’s mechanics are faithfully translated from the authors’ official MATLAB source into Python, preserving non-obvious implementation details — including the pairwise 2D rotation strategy for high-dimensional problems and the precomputed fractal figure bank (`LFND.mat`).

The algorithm is then adapted to the **Traveling Salesman Problem** (TSP) via a *Random Keys* encoding bridge. An ablation study across five TSPLIB instances (51–99 cities) quantifies the contribution of each architectural component: the raw continuous-to-discrete mapping produces tours 150–300% above optimal; adding 2-opt local search on the iteration-best agent reduces the mean gap to **6.84%**; and a double-bridge ILS escape mechanism further reduces it to **2.78%** on average, with a best-case gap of **0.000%** on berlin52.

This work demonstrates that while LA’s own update mechanism contributes minimal diversity in permutation space, its combination with classical local search and strategic perturbations yields competitive performance on benchmark instances with a fixed, untuned parameter set.

KEYWORDS

Lichtenberg Algorithm, Traveling Salesman Problem, Random Keys Encoding, 2-opt Local Search, Iterated Local Search, Double-Bridge Perturbation, Algorithm Reproduction, Combinatorial Optimization, Metaheuristics, TSPLIB Benchmarks.

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Introduction

This chapter establishes the project context, motivates the engineering problem of adapting a physics-inspired metaheuristic to a discrete combinatorial problem, and states the primary research objectives.

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1.1 Motivation

Reproducing published metaheuristics from source code is a technically demanding task that goes well beyond translating pseudocode. Source implementations often encode critical domain-specific choices — data normalization conventions, boundary handling semantics, hardcoded constants, and numerical precision decisions — that are invisible in the paper but materially affect results.

This project targets the `LICHTENBERG ALGORITHM (LA)`, a 2021 physics-inspired optimizer whose official MATLAB implementation embeds several such non-obvious decisions.

REPRODUCTION VS. RE-IMPLEMENTATION

Reproducing a published algorithm means matching the authors' code behavior, not just their equations. These are often not the same. Any constant, convention, or loop structure that differs between the paper's pseudocode and the released source must be identified, documented, and preserved in the reproduction.

1.2 Problem Statement

The **Traveling Salesman Problem** (TSP) asks: given n cities with known pairwise distances, find the shortest closed tour visiting each city exactly once. Formally:

$$\min_{\sigma \in S_n} \sum_{i=1}^n d(\sigma(i), \sigma(i \bmod n + 1)) \quad (1.1)$$

where S_n is the symmetric group of n cities and $d(\cdot, \cdot)$ is the Euclidean distance rounded to the nearest integer (TSPLIB `EUC_2D` convention).

* NP-HARDNESS OF TSP

TSP is NP-hard: no polynomial-time exact algorithm is known. For $n = 52$ cities, the number of candidate tours is $(52 - 1)!/2 \approx 10^{65}$ — larger than the number of atoms in the observable universe. Exact solvers (Concorde) exist but are impractical at scale; metaheuristics provide high-quality approximate solutions efficiently.

1.3 Objectives

This project pursues three engineering objectives:

1. **Faithful reproduction** of LA from the authors' official MATLAB source, including all non-documented implementation details.
2. **Adaptation** of LA to TSP via a Random Keys encoding bridge, and empirical evaluation of its effectiveness.
3. **Hybrid extension** with 2-opt and Iterated Local Search operators, validated through a three-configuration ablation study.

CENTRAL RESEARCH QUESTION

Can LA's internal exploration mechanism, when bridged to a discrete permutation space via Random Keys encoding, generate useful combinatorial search diversity? And if not, what is the minimal operator extension needed to make it competitive?

1.4 Report Structure

Chapter 2 covers the theoretical background on LA and TSP. Chapter 3 describes the engineering methodology and adaptation decisions. Chapter 4 provides the detailed implementation reference. Chapter 5 defines the experimental protocol. Chapter 6 presents and interprets the results. Chapter 7 offers a critical analysis. Chapter 8 concludes with future directions.

Background

This chapter presents the physical inspiration behind the Lichtenberg Algorithm, its core mechanics, and the TSPLIB benchmark instances used throughout this study.

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2.1 The Lichtenberg Algorithm

2.1.1 Physical Inspiration

The **Lichtenberg figure** is the branching, fractal discharge pattern formed when high-voltage electricity propagates through a dielectric material. This phenomenon, known as *Diffusion-Limited Aggregation* (DLA), produces a structure with a characteristic density profile: densely branched near the discharge origin, progressively sparser at the periphery.

Pereira et al. [1] recognized that this geometry encodes, in spatial form, the balance metaheuristics seek: concentrated local exploitation near the current best solution, and sparse exploratory branches reaching into distant regions.

2.1.2 Algorithm Mechanics

At each iteration, LA performs the following sequence:

1. **Anchor:** Set the trigger point $X_{\text{trigger}} \leftarrow X_{\text{best}}$.
2. **Rotate:** Apply a random 2D rotation $R(\theta)$ to the fractal cloud. For dimensions $d > 3$, pairs of coordinates are rotated independently.
3. **Scale:** Multiply by $s_f = 1.2 \times \text{rand}()$.
4. **Recenter:** Shift the cloud so its centroid aligns with X_{trigger} .
5. **Evaluate & Replace:** Each agent picks a candidate from the cloud and applies greedy acceptance.

PRECOMPUTED FIGURE BANK

For the $M = 0$ mode (used throughout this project), the fractal cloud is not regenerated at runtime. The authors provide a precomputed figure (`LFND.mat`: 7,792 points in $\mathbb{R}^2$, raw pixel coordinates) that is reused every iteration. The cloud is not pre-centered: the pipeline's own recenter step handles coordinate alignment.

2.2 TSPLIB Benchmark Instances

All experiments use standard TSPLIB instances [2]. Table 2.1 summarizes the five instances used in the scaling study.

Table 2.1. TSPLIB benchmark instances used in this study.

Instance	Origin	Cities	Optimum	Type
eil51	Christofides & Eilon	51	426	EUC_2D
berlin52	Groetschel (Berlin)	52	7,542	EUC_2D
st70	Smith & Thompson	70	675	EUC_2D
pr76	Padberg & Rinaldi	76	108,159	EUC_2D
rat99	Pulleyblank	99	1,211	EUC_2D

▲ DISTANCE CONVENTION

TSPLIB defines edge weights as $d_{ij} = \text{nint}\left(\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}\right)$ — Euclidean distance *rounded to the nearest integer*. Using raw floating-point distances produces results that cannot be compared against published optima. For `pr76`, the difference is hundreds of units relative to the 108,159 optimum.

Methodology

This chapter describes the Random Keys encoding bridge between LA's continuous output and the TSP permutation space, and presents the complete hybrid LA-ILS architecture.

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3.1 Encoding Bridge: Random Keys

* RANDOM KEYS ENCODING

Each agent is a vector $\mathbf{k} \in [0, 1]^n$. The decoded tour is $\sigma = \text{argsort}(\mathbf{k})$: the city with the smallest key is visited first. This mapping is surjective over S_n and produces a valid tour for any real-valued input — no repair mechanism is needed.

The encoding bridge must satisfy three properties to be compatible with LA's continuous update rule:

- **Surjectivity:** every valid tour must be reachable from some key vector.
- **Automatic feasibility:** no repair mechanism needed; any vector in $[0, 1]^n$ decodes to a valid permutation.
- **Preservation of LA's mechanics:** the rotation, scaling, and recentering operations in LA remain well-defined on the continuous key vector.

⚠ INFORMATION LOSS VIA ARGSORT

The `argsort` decoding preserves only *ordinal rank* of the keys — all magnitude information from LA's fractal cloud is discarded. Two key vectors at Euclidean distance ϵ can decode to completely different tours if any pair of keys are near-tied. LA's rotation and scaling operations, designed to exploit continuous proximity, have no meaningful effect on permutation quality through this decoding.

3.2 Hybrid Architecture

The complete LA-ILS pipeline combines four components with non-overlapping responsibilities:

Algorithm 1. LA-ILS Hybrid for TSP

Input: TSPLIB instance (n cities), Pop, N_{iter} , ref, restart_patience

Output: Best tour σ^* , best length f^*

```

1 Load fractal cloud  $\mathcal{F}$  from LFND.mat;
2 Initialize Pop random key vectors  $\mathbf{k}_i \in [0, 1]^n$ ;
3 foreach agent  $i$  do
4    $\sigma_i \leftarrow \text{argsort}(\mathbf{k}_i)$ ;  $f_i \leftarrow \text{tour\_length}(\sigma_i)$ ;
5  $f^* \leftarrow \min_i f_i$ ;  $\mathbf{k}^* \leftarrow \arg \min_i f_i$ ; anchor  $\leftarrow \mathbf{k}^*$ ; stag  $\leftarrow 0$ ;
6 for  $t = 1$  to  $N_{\text{iter}}$  do
7    $s_f \leftarrow 1.2 \times \text{rand}()$ ;
8    $\mathcal{X}_{\text{global}} \leftarrow \text{la\_points}(\mathcal{F}, [0, 1]^n, \text{anchor}, s_f, n)$ ;
9    $\mathcal{X}_{\text{local}} \leftarrow \text{la\_points}(\mathcal{F}, [0, \text{ref}]^n, \text{anchor}, s_f, n)$ ;
10  foreach agent  $i$  do
11     $\mathbf{k}'_i \leftarrow \text{sample from } \mathcal{X}_{\text{global}} (60\%) \text{ or } \mathcal{X}_{\text{local}} (40\%);$ 
12    if  $\text{tour\_length}(\text{argsort}(\mathbf{k}'_i)) \leq f_i$  then replace agent  $i$ ;
13    if  $f_i \leq f^*$  then  $f^* \leftarrow f_i$ ;  $\mathbf{k}^* \leftarrow \mathbf{k}_i$ ; anchor  $\leftarrow \mathbf{k}_i$ ;
    // Memetic step: 2-opt on iteration-best only
14     $\sigma_{\text{best}} \leftarrow \text{2-opt}(\text{argsort}(\mathbf{k}^*))$ ;
15    if  $\text{len}(\sigma_{\text{best}}) < f^*$  then update  $f^*$ ,  $\mathbf{k}^*$ , anchor;
    // Stagnation escape
16    if  $f^*$  improved then
17      stag  $\leftarrow 0$ 
18    else
19      stag  $\leftarrow \text{stag} + 1$ 
20    if stag  $\geq \text{restart\_patience}$  then
21      anchor  $\leftarrow \text{encode}(\text{2-opt}(\text{double\_bridge}(\sigma^*)))$ ;
22      Reinitialize worst 20% of population uniformly;
23      stag  $\leftarrow 0$ ;
24 return  $\text{argsort}(\mathbf{k}^*), f^*$ 

```

ANCHOR VS. BEST-EVER SEPARATION

`best_ever` tracks the globally best solution and is *never* degraded. `anchor` is the cloud recentering point and is *allowed to wander* after a double-bridge kick, even if the kick does not immediately improve the

objective. This is what enables genuine basin escape: the search center relocates before a new region proves better.

Implementation

This chapter provides a detailed technical reference for the Python implementation, covering file structure, critical MATLAB-to-Python translation decisions, and the modular hook mechanism.

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4.1 File Architecture

Table 4.1. Source file responsibilities.

File	Responsibility
<code>la_optimization.py</code>	Faithful Python translation of <code>LA_optimization.m</code> . Owns all LA mechanics: figure loading, pairwise rotation, scaling, recentering, greedy replacement, stagnation tracking. Exposes two hooks: <code>two_opt_hook</code> and <code>perturb_hook</code> .
<code>tsp_problem.py</code>	TSP domain layer. Contains all five TSPLIB instance coordinates, EUC_2D distance matrix, Random Keys decoder, 2-opt operator, double-bridge, and tour re-encoder.
<code>MAIN.py</code>	Single-instance runner for berlin52. Wires hooks, runs one experiment, plots convergence and the best tour.
<code>benchmark.py</code>	Full ablation study: 3 configurations $\times$ 5 instances $\times$ 5 seeds = 75 independent runs.

4.2 Critical Translation Decisions

4.2.1 Pairwise 2D Rotation ($d > 3$)

⚠ ODD-DIMENSION ALLOCATION BUG

In MATLAB, the loop `for i=1:2:d+2` auto-expands the output matrix for odd d . Python's pre-allocated arrays do not. This caused a silent crash on `eil51` ($d = 51$) that did not appear on `berlin52` ($d = 52$). The fix: allocate width $d + 3$ for odd d , $d + 2$ for even.

```
1 # MATLAB: for i = 1:2:d+2 -- auto-grows when d is odd
2 # Python: must pre-allocate the correct width explicitly
3 width = d + 2 if d % 2 == 0 else d + 3
4 Xi = np.zeros((Np, width))
5 for i in range(0, width, 2):
6     gama = rng.random()
7     c, s = np.cos(gama), np.sin(gama)
8     R = np.array([[c, -s], [s, c]])
9     Xi[:, i:i+2] = (R @ K.T).T
10 Xi = Xi[:, :d] # trim to d columns
```

Listing 4.1. Pairwise rotation with correct odd-dimension width.

4.2.2 Hardcoded Population Split

📌 IMPLEMENTATION NOTE

In `LA_optimization.m`, line 18 reads `pop1 = round((0.4)*n)`. This is a *literal constant*, not derived from the user-supplied `ref` parameter (which only rescales the cloud bounds). It is preserved verbatim.

```
1 # NOTE: 0.4 is a literal constant in LA_optimization.m line 18.
2 # ref only controls cloud scale, NOT this split.
3 pop1 = int(round(0.4 * n)) # local agents (40%)
4 pop2 = n - pop1          # global agents (60%)
```

Listing 4.2. Preserving the authors' hardcoded 0.4 constant.

4.2.3 TSPLIB Distance Convention

```
1 @staticmethod
2 def _build_distance_matrix(coords: np.ndarray) -> np.ndarray:
3     """
```

```

4     TSPLIB EUC_2D: d_ij = nint(sqrt((xi-xj)^2 + (yi-yj)^2))
5     CRITICAL: must use np.round(), not raw float.
6     """
7     diff = coords[:, None, :] - coords[None, :, :]
8     raw = np.sqrt((diff ** 2).sum(axis=-1))
9     return np.round(raw)

```

Listing 4.3. TSPLIB-compliant EUC_2D distance matrix.

4.3 Hook Mechanism

```

1 def optimize(
2     self,
3     verbose: bool = True,
4     two_opt_hook=None,      # callable(keys) -> (keys, fitness)
5     perturb_hook=None,     # callable(keys, rng) -> (keys, fitness)
6     restart_patience: int = 6,
7     restart_fraction: float = 0.2,
8 ):
9     """
10    two_opt_hook : applied to iteration-best every iteration.
11    perturb_hook  : applied to best_ever on stagnation trigger.
12    la_optimization.py knows nothing about cities or permutations.
13    All TSP logic is injected by the caller.
14    """

```

Listing 4.4. Hook signature — keeps la_optimization.py domain-agnostic.

Experimental Setup

This chapter defines the ablation protocol and the fixed hyperparameter budget used identically across all instances and configurations.

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Ablation Protocol

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5.1 Ablation Protocol

Three configurations are evaluated under an identical budget to isolate the contribution of each architectural component:

Table 5.1. Three ablation configurations — identical budget, single variable changed.

Configuration	2-opt	Double-Bridge	Restarts
Pure LA + Random Keys	×	×	×
LA + 2-opt (No Kicks)	✓	×	×
Full Hybrid LA-ILS	✓	✓	✓

5.2 Hyperparameters

Table 5.2. Fixed hyperparameters — identical across all instances and configurations.

Parameter	Value
Population size	50
Iterations	200
Refinement (<code>ref</code>)	0.4
2-opt passes	60 (runs to local convergence)
Stagnation patience	6 iterations
Restart fraction	20% of population
Seeds per instance	5 (seeds 42–46)
Figure bank	<code>LFND.mat</code> (authors' official, $M=0$)
Distance convention	TSPLIB EUC_2D (rounded integer)

▣ WHY 5 SEEDS?

A single seed is not a valid performance estimate for a stochastic algorithm with high trajectory variance. Five seeds provides a first-order variance estimate. For publication-grade benchmarks, 20–30 seeds are standard.

Results

This chapter presents the ablation study results and the full hybrid performance across five TSPLIB instances and five random seeds.

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6.1 Ablation Study

Table 6.1 presents mean optimality gap over 5 seeds for each configuration.

Table 6.1. Ablation study: mean optimality gap (%) over 5 seeds, three configurations.

Instance	Cities	Pure LA+RK	LA+2-opt	Full Hybrid	2-opt Gain	Kicks Gain
eil51	51	186.95%	5.87%	2.54%	+181.1 pp	+3.3 pp
berlin52	52	188.87%	6.69%	2.68%	+182.2 pp	+4.0 pp
st70	70	303.82%	8.06%	2.96%	+295.8 pp	+5.1 pp
pr76	76	159.92%	3.93%	2.00%	+156.0 pp	+1.9 pp
rat99	99	176.32%	9.64%	3.73%	+166.7 pp	+5.9 pp
Average	—	203.18%	6.84%	2.78%	+196.3 pp	+4.1 pp

* ABLATION FINDING

97% of the total error reduction (203.18% → 6.84%) comes from adding 2-opt to the iteration-best agent alone. The double-bridge kick contributes a further 4.1 percentage points. **LA's physics-inspired update mechanism does not generate useful combinatorial diversity through `argsort` decoding.**

6.2 Full Hybrid Performance

Table 6.2. Full Hybrid LA-ILS performance over 5 seeds per instance.

Instance	Cities	Optimum	Mean Gap (%)	Std (%)	Best Gap (%)	Time (s)
eil51	51	426	2.535	0.908	0.939	4.34
berlin52	52	7,542	2.684	2.691	0.000	4.32
st70	70	675	2.963	1.442	1.185	5.63
pr76	76	108,159	2.001	0.917	0.488	6.27
rat99	99	1,211	3.732	2.313	0.578	9.94
Average	—	—	2.783	1.654	0.638	6.10

BERLIN52 EXACT OPTIMUM

Seed 42 achieves a **0.000%** gap on berlin52 — the algorithm finds the exact global optimum (7,542). However, the standard deviation of 2.691% on the same instance reveals that this is seed-sensitive, not a guaranteed behavior.

6.3 Convergence Analysis

All five instances display the same qualitative convergence profile: rapid initial improvement in the first 5–10 iterations as LA fills the population with diverse starting points, followed by extended plateaus broken by occasional ILS kicks. The staircase pattern visible in the convergence plots is the fingerprint of an iterated local search operating correctly — each sudden drop corresponds to a double-bridge perturbation successfully escaping a 2-opt local optimum.

Critical Analysis

This chapter critically examines the strengths and limitations of the LA-ILS framework, including computational complexity and scalability.

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7.1 Strengths

- **Faithful reproduction.** Every internal mechanic of the original MATLAB implementation is preserved, including non-documented behaviors: elitist row injection, hardcoded population split, and unsigned pixel coordinates in `LFND.mat`.
- **Modular hook design.** `la_optimization.py` is entirely domain-agnostic. Adapting the framework to a different combinatorial problem requires only changing the hook functions.
- **Statistically sound evaluation.** Results are reported as mean $\pm$ standard deviation over five seeds. No cherry-picked single-run numbers.
- **Competitive performance.** A 2.78% mean gap on instances up to 99 cities in under 10 seconds, with a fixed untuned hyperparameter set.

7.2 Weaknesses

▲ LA'S MARGINAL CONTRIBUTION

The ablation establishes that LA's own update mechanism contributes almost nothing to solution quality on TSP. The algorithm's geometric operations — rotation, scaling, recentering — operate on continuous key

vectors whose ordinal ranking (the only information `argsort` preserves) has no meaningful correlation with tour quality. LA effectively functions as a random population initializer in this framework.

▲ RANDOM KEYS INFORMATION LOSS

The `argsort` decoding discards all magnitude information from LA's carefully structured fractal cloud. Two key vectors at Euclidean distance ϵ in $[0, 1]^n$ can decode to completely different tours if any pair of keys are near-tied. LA was designed to exploit continuous proximity; Random Keys systematically destroys it.

- **High seed variance.** Standard deviations reach the same order of magnitude as mean gaps (berlin52: mean 2.684%, std 2.691%).
- **Platform sensitivity.** Trigonometric functions (`sin/cos`) differ by last-bit between Linux glibc and Windows MSVC, producing materially different search trajectories from the same seed — observed on `eil51` and `rat99`.

7.3 Computational Complexity

Table 7.1. Per-iteration computational cost of each component.

Component	Per-Iteration Cost	Dominant?
LA cloud generation	$\mathcal{O}(N_p \cdot d)$, $N_p = 7792$	Moderate
Population evaluation	$\mathcal{O}(n_{\text{pop}} \cdot n)$	Low
2-opt (best agent)	$\mathcal{O}(n^2)/\text{pass}$, 60 passes max	Low ($n \leq 100$)
Double-bridge + 2-opt	$\mathcal{O}(n^2)$, amortized over 6 iters	Negligible

▣ SCALABILITY THRESHOLD

At $n = 99$, $\mathcal{O}(n^2) = 9801$ operations per 2-opt pass — trivial. For $n > 500$, the 60-pass 2-opt becomes the dominant cost and the best-agent-only strategy would need revisiting. The Random Keys landscape also becomes increasingly high-dimensional, making it vanishingly unlikely that any LA candidate produces a better permutation ranking.

Conclusion

This chapter summarizes the three engineering contributions of this project and outlines realistic future directions for the framework.

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8.1 Summary of Contributions

This project delivers three distinct engineering contributions:

1. **A faithful Python reproduction** of the Lichtenberg Algorithm, validated against the authors' official MATLAB source. All internal mechanics — precomputed fractal figure bank, pairwise 2D rotation with correct odd-dimension allocation, elitist row injection, hardcoded population split — are preserved and documented.
2. **An empirically grounded adaptation** to TSP via Random Keys encoding, with a rigorous three-configuration ablation study across five TSPLIB instances and five seeds each. Every component's contribution is quantified independently.
3. **An honest characterization of LA's limits** in combinatorial optimization. LA's physics-inspired update mechanism does not translate useful geometric information through `argsort` decoding. Solution quality in the final hybrid is driven by classical 2-opt and ILS operators.

* FINAL PERFORMANCE

2.78% mean optimality gap across five TSPLIB instances (51–99 cities) in under 10 seconds per run, with a best-case gap of **0.000%** on berlin52. Achieved with a fixed, untuned hyperparameter set and no instance-specific tuning.

8.2 Future Work

Alternative discrete encodings. Replace Random Keys with a greedy construction heuristic biased by the continuous key vector. This would restore the geometric meaning of LA's rotation and scale operations.

Adaptive stagnation patience. Replace the fixed `restart_patience = 6` with a population-diversity criterion (e.g., variance of Kendall τ distances between decoded tours).

Lin-Kernighan moves. Replace double-bridge with a Lin-Kernighan style k -opt kick, which is empirically more effective at escaping 2-opt basins.

Parallel multi-start. Run 8–16 independent LA-ILS instances on a multi-core CPU, sharing `best_ever` across threads to improve solution quality within the same wall-clock budget.

GPU acceleration. The cloud generation matrix operations in `la_points` are naturally vectorizable. A CuPy port would reduce per-iteration cloud generation cost by 10–100 $\times$ on modern hardware.

Reinforcement-learning operator selection. Train an RL agent to decide, per iteration, which local search depth to apply based on current stagnation state and population statistics — replacing the fixed 60-pass 2-opt with an adaptive schedule learned across a training distribution of TSP instances.

Code: All source files available upon request. Dependencies: Python 3.12, NumPy, SciPy, Matplotlib. **Ref:** Pereira et al. (2021). *Expert Systems with Applications*, 170, 114522.

Hyperparameter Reference

In this chapter

A.1 Complete Table

A.2 Sensitivity Notes

Complete reference table for all hyperparameters used across every experiment in this report, with source attribution and sensitivity notes.

A.1 Complete Hyperparameter Table

Table A.1. All hyperparameters used across every experiment.

Parameter	Value	Source	Notes
Population size	50	Authors (MAIN.m)	$10 \times d$ in paper
Iterations	200	This work	Fixed budget
<code>ref</code>	0.4	Authors	Cloud scale factor
<code>M</code>	0	Authors	Load precomputed figure
2-opt max passes	60	This work	Converges in practice
Stagnation patience	6	This work	Tuned empirically
Restart fraction	20%	This work	Of population per kick
Seeds	42–46	This work	5 independent runs
Distance convention	EUC_2D	TSPLIB standard	<code>np.round()</code>
Figure bank points	7,792	<code>LFND.mat</code>	Authors' precomputed

A.2 Sensitivity Notes

☐ STAGNATION PATIENCE

The `restart_patience = 6` value was selected after observing that the algorithm typically plateaus within 5–8 iterations of entering a 2-opt-local optimum. Values below 4 cause excessive perturbation preventing convergence; values above 12 waste most of the iteration budget. This parameter is the most impactful tuning variable.

MATLAB-to-Python Translation

In this chapter

B.1 Function Correspondence

B.2 Non-Obvious Behaviors

Function correspondence table and documentation of non-obvious MATLAB behaviors preserved in the Python reproduction.

B.1 Function Correspondence

Table B.1. Correspondence between MATLAB source functions and Python equivalents.

MATLAB	Python	Notes
<code>LA_figure(d, Np, Rc, S, 0)</code>	<code>load_LF("LFND.mat")</code>	M=0: load precomputed. Returns raw ($N_p, 2$) uint16, uncentered.
<code>LA_points(K, LB, UB, x0, sf, d)</code>	<code>la_points(K, lb, ub, x0, sf, d, rng)</code>	Added <code>rng</code> for reproducibility.
<code>bound_check(s, LB, UB)</code>	<code>bound_check(s, lb, ub)</code>	<code>np.clip(s, lb, ub)</code> — identical semantics.
<code>for i=1:2:d+2</code>	<code>range(0, width, 2)</code>	Width is $d+2$ (even d) or $d+3$ (odd d).
<code>round(Np/2)</code>	<code>int(round(Np/2.0))-1</code>	MATLAB 1-indexed; Python 0-indexed. Off-by-one adjusted.

B.2 Non-Obvious Behaviors Preserved

🔗 ELITIST ROW INJECTION

After scaling, MATLAB overwrites row `round(Np/2)` with the cloud centroid P_{cc} , then shifts the entire cloud by $\delta = P_{cc} - X_0$. The net effect: one specific row always equals X_0 exactly after the shift — a built-in elitism not mentioned in the paper. Preserved as `Xi[mid, :] = Pcc` before the delta shift.

🔗 HARDCODED 0.4/0.6 SPLIT

Despite `ref` being a user parameter, the population split is `pop1 = round(0.4 * n)` regardless of `ref`'s value. `ref` only scales the local cloud's bounds (`LB*ref` to `UB*ref`). The split ratio is independent. Preserved verbatim.

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- [1] J. L. J. Pereira, M. B. Francisco, C. A. Diniz, G. A. Oliver, S. S. Cunha Jr., and G. F. Gomes, "Lichtenberg algorithm: A novel hybrid physics-based meta-heuristic for global optimization," *Expert Systems with Applications*, vol. 170, p. 114522, 2021.
- [2] G. Reinelt, "TSPLIB — A traveling salesman problem library," *ORSA Journal on Computing*, vol. 3, no. 4, pp. 376–384, 1991.